Week 2 — Data Cleaning, Statistics & Exploratory Data Analysis

Industrial Training TR-103 · Data Analytics Organisation: World With Web **Trainee:** Savreet Kaur · URN 2302668 **Period:** 22 June 2026 – 26 June 2026

Objective of the week

Week 1 ended with loading a dataset and seeing its problems. This week is about fixing those problems and then describing what remains: missing values, duplicates and outliers first, then descriptive statistics, relationships between variables, and a complete exploratory analysis.

Day 6 — Monday, 22 June 2026

Missing values

Three questions must be answered before anything is changed: how much is missing, where, and why.

```
df.isnull().sum()                    # count per column
df.isnull().mean().mul(100).round(1)   # percentage per column
```

Handling strategies and when each applies:

STRATEGY	CODE	APPROPRIATE WHEN
Drop rows	`df.dropna()`	Few rows affected, data is plentiful
Drop column	`df.drop(columns=[...])`	Column is mostly empty and not essential
Fill constant	`df.fillna(0)`	Missing genuinely means zero
Fill mean	`df["x"].fillna(df["x"].mean())`	Roughly symmetric numeric data
Fill median	`df["x"].fillna(df["x"].median())`	Skewed data, or outliers present
Forward fill	`df.ffill()`	Time series where the last value persists

The point stressed repeatedly was that **filling is an assumption**. Replacing missing salaries with the mean does not recover the real values; it makes the column usable while reducing its apparent variance. Any such choice should be recorded.

Duplicates

```
df.duplicated().sum()
df = df.drop_duplicates()
df = df.drop_duplicates(subset=["order_id"], keep="first")
```

The distinction between a fully duplicated row and a repeated key was covered — the second often indicates a genuine data-entry problem rather than a harmless repeat.

Data types

Numeric columns read as text are the most common silent defect, usually caused by stray symbols:

```
df["amount"] = (df["amount"]
                .str.replace(",", "", regex=False)
                .str.replace("₹", "", regex=False)
                .astype(float))

df["order_date"] = pd.to_datetime(df["order_date"], errors="coerce")
```

`errors="coerce"` turns unparseable dates into `NaT` rather than raising, so the bad rows can be counted and inspected instead of stopping the script.

Takeaway

Cleaning is not a preliminary step to the analysis — the decisions made here determine what the analysis is able to say.

Day 7 — Tuesday, 23 June 2026

Measures of central tendency

MEASURE	MEANING	SENSITIVE TO OUTLIERS
Mean	Arithmetic average	Yes
Median	Middle value when sorted	No
Mode	Most frequent value	No

The worked example made the difference concrete: in a salary column with a few very high values, the mean sat above the median, and the mean described almost nobody in the dataset. The rule given was to report the median for skewed data and to state which was used.

Measures of spread

- **Range** — maximum minus minimum; entirely determined by two points
- **Variance** — mean squared deviation from the mean
- **Standard deviation** — square root of variance, in the original units
- **IQR** — Q3 minus Q1, the middle 50 per cent of the data

```
df["salary"].describe()

q1, q3 = df["salary"].quantile([0.25, 0.75])
iqr = q3 - q1
```

Takeaway

Standard deviation is readable because it is in the same unit as the data; variance is not. Both describe the same thing.

Day 8 — Wednesday, 24 June 2026

Distributions

Histograms and box plots were used to see the shape of a column rather than only its summary numbers.

- **Normal** — symmetric, mean ≈ median
- **Right-skewed** — a long tail of high values; mean > median. Income, prices and web traffic are typically this shape.
- **Left-skewed** — a long tail of low values; mean < median

Outlier detection

IQR method — anything outside 1.5 × IQR beyond the quartiles:

```
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
outliers = df[(df["salary"] < lower) | (df["salary"] > upper)]
```

Z-score method — anything more than 3 standard deviations from the mean. Noted as unreliable on skewed data, because the mean and standard deviation used to compute it are themselves distorted by the outliers.

What to do with an outlier

Four options were discussed, in order of preference: investigate it, keep it, cap it (winsorise), or remove it. Removing should be the last resort and must be recorded — a genuine extreme value is data, and deleting it to make a chart look tidier is falsifying the result.

Takeaway

An outlier is not automatically an error. A data-entry mistake and a genuinely exceptional customer look identical in the numbers; only context separates them.

Day 9 — Thursday, 25 June 2026

Correlation

```
df[["sales", "profit", "discount"]].corr()
```

Pearson's correlation coefficient runs from -1 to +1 and measures the strength of a **linear** relationship.

RANGE	INTERPRETATION
0.7 to 1.0	Strong positive
0.3 to 0.7	Moderate positive
-0.3 to 0.3	Weak or none
-1.0 to -0.7	Strong negative

Two cautions were emphasised. A coefficient near zero rules out a *linear* relationship, not any relationship — a clear curve can score near zero. And correlation does not imply causation; the standard counter-example of ice-cream sales and drowning incidents, both driven by temperature, was used.

Introduction to hypothesis testing

- **Null hypothesis (H_0)** — there is no effect or no difference
- **Alternative hypothesis (H_1)** — there is one
- **p-value** — the probability of seeing a result at least this extreme if H_0 were true

- **Significance level (α)** — conventionally 0.05

A two-sample t-test was run comparing the mean sales of two regions:

```
from scipy import stats

north = df[df["region"] == "North"]["sales"]
south = df[df["region"] == "South"]["sales"]
t_stat, p_value = stats.ttest_ind(north, south, equal_var=False)
```

The interpretation was stated carefully: a p-value below 0.05 means the data would be unlikely under the null hypothesis. It does not give the probability that the null hypothesis is true, and it says nothing about whether the difference is large enough to matter in practice.

Takeaway

Statistical significance and practical importance are different questions. With a large enough sample, a trivial difference becomes significant.

Day 10 — Friday, 26 June 2026

Grouping and aggregation

```
df.groupby("region")["sales"].sum()

df.groupby(["region", "category"]).agg(
    total_sales=("sales", "sum"),
    avg_profit=("profit", "mean"),
    orders=("order_id", "count"),
).reset_index()
```

The split–apply–combine idea was explained as the direct counterpart of SQL's `GROUP BY`, which Week 3 covers.

Pivot tables

```
pd.pivot_table(df, index="region", columns="category",
               values="sales", aggfunc="sum", fill_value=0)
```

Full EDA on a sample dataset

The week closed with an end-to-end exploratory analysis of a retail sales dataset, following a fixed sequence:

1. Load and inspect — shape, types, nulls
2. Clean — fix types, handle missing values, remove duplicates
3. Univariate analysis — distribution of each important column
4. Bivariate analysis — relationships between pairs of columns
5. Grouped summaries — by region, category and month
6. Write down the findings in plain sentences

Findings recorded

- Sales are right-skewed: most orders are small, a few are very large
- One region accounts for a disproportionate share of total revenue

- Discount and profit are negatively correlated, with profit turning negative above roughly a 30 per cent discount
- Monthly sales show a clear year-end peak

Takeaway

Exploratory analysis is for forming questions, not confirming conclusions. Each finding above is a hypothesis to test, not a result to report as settled.

Summary of Week 2

DAY	DATE	TOPIC
6	Mon 22 Jun	Missing values, duplicates, data types
7	Tue 23 Jun	Central tendency and spread
8	Wed 24 Jun	Distributions, skewness, outliers
9	Thu 25 Jun	Correlation and hypothesis testing
10	Fri 26 Jun	Group-by, pivot tables, complete EDA

Skills gained: cleaning a messy dataset and documenting the choices · describing data with the right summary statistic · detecting outliers · reading a correlation matrix · running and interpreting a t-test · performing a structured exploratory analysis.

Next week: the same questions, answered against a database using SQL.